


```
gvs_3@pye:~/t6$ python3 3b_socket-http-server.example.py 8403
gvs_3 - Server 2 HTTP manual pornit pe portul 8403
curl< pentru oprire.
```

```
gvs_3@pye:~/t6$ curl -v http://10.10.10.10:8403/
* IP=1: (none)
* IP=4: 37.120.249.45
* Trying 37.120.249.45:8403...
* Connected to sys.as.ro:8403 (37.120.249.45) port 8403
* GET / HTTP/1.1
* Host: sys.as.ro:8403
* User-Agent: curl/8.5.0
* Accept: */*
>
< HTTP/1.1 200 OK
< Server: nginx/1.29.7
< Date: Fri, 27 Mar 2026 13:27:55 GMT
< Content-Type: text/html
< Content-Length: 274
< Connection: keep-alive
< X-Student-Lab: Seminar6 - gvs_3
<
< [DOCTYPE html]
<html>
<head>
  <meta charset="utf-8">
  <title>Server HTTP Manual</title>
</head>
<body>
  <div>gvs_3 Server 2 HTTP implementat cu socket-curl</div>
  <p>Acesta este pagina servita Manual din directorul <code>static</code>.</p>
</body>
</html>
* Connection #0 to host sys.as.ro left intact
```

```

git_@kali:~$ sudo netstat -anpt | grep 58403
(Not all processes could be identified, non-owned process info
will not be shown, you would have to be root to see it all.)
tcp        0      0 0.0.0.0:0.0.0.0:58403    0.0.0.0:*
LISTEN     327680/*nix: mast
git_@kali:~$ sudo netstat -anpt | grep 58403
(Not all processes could be identified, non-owned process info
will not be shown, you would have to be root to see it all.)
tcp        0      0 0.0.0.0:0.0.0.0:58403    0.0.0.0:*
tcp        0      0 37.120.249.45:5152714    37.120.249.45:58403
LISTEN     327680/*nix: mast
TIME_WAIT -
git_@kali:~$ sudo netstat -anpt | grep 8403
(Not all processes could be identified, non-owned process info
will not be shown, you would have to be root to see it all.)
tcp        0      0 0.0.0.0:0.0.0.0:8403     0.0.0.0:*
LISTEN     327680/*nix: mast
tcp        0      0 37.120.249.45:52714      37.120.249.45:8403
TIME_WAIT -
git_@kali:~$ sudo netstat -anpt | grep 8403
(Not all processes could be identified, non-owned process info
will not be shown, you would have to be root to see it all.)
tcp        0      0 0.0.0.0:0.0.0.0:8403     0.0.0.0:*
LISTEN     327680/*nix: mast
tcp        0      0 37.120.249.45:39640      37.120.249.45:8403
TIME_WAIT -
git_@kali:~$

```

```

gr5_3@sys: ~
gr5_3@sys:~$ curl http://example.com >> curl_basics_log.txt
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
100  528      0  528    0     0   6331      0 --:--:-- --:--:-- --:--:--   6361
gr5_3@sys:~$ curl -v http://example.com >> curl_basics_log.txt
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--    0*
Host example.com:80 was resolved.
* IPv6: 2606:4700::6812:1a78, 2606:4700::6812:1b78
* IPv4: 104.18.27.120, 104.18.26.120
* Trying 104.18.27.120:80...
* Connected to example.com (104.18.27.120) port 80
> GET / HTTP/1.1
> Host: example.com
> User-Agent: curl/8.5.0
> Accept: */*
>
< HTTP/1.1 200 OK
< Date: Thu, 02 Apr 2026 16:17:48 GMT
< Content-Type: text/html
< Transfer-Encoding: chunked
< Connection: keep-alive
< Server: cloudflare
< Last-Modified: Tue, 24 Mar 2026 22:07:32 GMT
< Allow: GET, HEAD
< Accept-Ranges: bytes
< Age: 9962
< cf-cache-status: HIT
< CF-RAY: 9e6125547976c9de-OTP
<
{ [528 bytes data]
100  528      0  528    0     0  58041      0 --:--:-- --:--:-- --:--:--  66000
* Connection #0 to host example.com left intact
gr5_3@sys:~$ curl -I http://example.com >> curl_basics_log.txt
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--    0
gr5_3@sys:~$ mc

gr5_3@sys:~$ █

```

1.

De ce este util să vezi request-ul complet?

Pentru a verifica exact header-ele și datele trimise și a identifica rapid erori sau parametri lipsă.

De ce curl este preferat browserului?

Pentru că oferă control total asupra cererii și afișează datele brute, fără interferențe (cache, extensii).

2.

```
gr5_3@sys: ~/s6
gr5_3@sys:~$ cd s6/
gr5_3@sys:~/s6$ python3 2b_simple-http-builtin_example.py
Utilizare: python3 simple_http_builtin.py <port>
gr5_3@sys:~/s6$ python3 2b_simple-http-builtin_example.py 8000
Salut-gr5_3- Pornim serverul pe portul 8000...
Server pornit. Ctrl+C pentru oprire.
127.0.0.1 - - [02/Apr/2026 19:22:14] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [02/Apr/2026 19:22:25] "GET /hello HTTP/1.1" 200 -
127.0.0.1 - - [02/Apr/2026 19:22:34] "GET /api/time HTTP/1.1" 200 -
^C
Opreste server...
gr5_3@sys:~/s6$ python3 2b_simple-http-builtin_example.py 8000
File "/home/an3/gr5_3/s6/2b_simple-http-builtin_example.py", line 59
    elif self.path == "/student"
    ^
SyntaxError: expected ':'
gr5_3@sys:~/s6$ python3 2b_simple-http-builtin_example.py 8000
Salut-gr5_3- Pornim serverul pe portul 8000...
Server pornit. Ctrl+C pentru oprire.
127.0.0.1 - - [02/Apr/2026 19:29:33] "GET /student HTTP/1.1" 200 -
^C

gr5_3@sys:~/s6$
gr5_3@sys:~/s6$ GET /api/time HTTP/1.1
> Host: localhost:8000
> User-Agent: curl/8.5.0
> Accept: */*
>
* HTTP 1.0, assume close after body
< HTTP/1.0 200 OK
< Server: SimpleHTTP/0.6 Python/3.12.3
< Date: Thu, 02 Apr 2026 16:22:34 GMT
< Content-Type: application/json
< Content-Length: 73
<
{ [73 bytes data]
100 73 100 73 0 0 42147 0 --:--:-- --:--:-- --:--:-- 73000
* Closing connection
gr5_3@sys:~/s6$ mc
gr5_3@sys:~/s6$ mc
gr5_3@sys:~/s6$ curl -v http://localhost:8000/student >> builtin_http_log.txt
* Host localhost:8000 was resolved.
* IPv6: ::1
* IPv4: 127.0.0.1
% Total % Received % Xferd Average Speed Time Time Time Current
0 0 0 0 0 0 0 0 --:--:-- --:--:-- --:--:-- 0* Trying [::1]:8000...
* connect to ::1 port 8000 from ::1 port 37270 failed: Connection refused
* Trying 127.0.0.1:8000...
* Connected to localhost (127.0.0.1) port 8000
> GET /student HTTP/1.1
> Host: localhost:8000
> User-Agent: curl/8.5.0
> Accept: */*
>
* HTTP 1.0, assume close after body
< HTTP/1.0 200 OK
< Server: SimpleHTTP/0.6 Python/3.12.3
< Date: Thu, 02 Apr 2026 16:29:33 GMT
< Content-Type: application/json
< Content-Length: 39
<
{ [39 bytes data]
100 39 100 39 0 0 19042 0 --:--:-- --:--:-- --:--:-- 39000
* Closing connection
gr5_3@sys:~/s6$
```

3.

 gr5_3@sys: ~/s6

```

00...
* connect to ::1 port 8000 from ::1 port 60534 failed: Connection refused
* Trying 127.0.0.1:8000...
* Connected to localhost (127.0.0.1) port 8000
> GET /index.html HTTP/1.1
> Host: localhost:8000
> User-Agent: curl/8.5.0
> Accept: */*
>
< HTTP/1.1 200 OK
< Content-Type: text/html
< Content-Length: 271
< Connection: close
<
{ [271 bytes data]
100 271 100 271 0 0 230k 0 --:--:-- --:--:-- --:--:-- 264k
* Closing connection
<!DOCTYPE html>
<html>
<head>
<meta charset="utf-8">
<title>Server HTTP Manual</title>
</head>
<body>
<h1>gr5_3 Server 2 HTTP implementat cu socket-uri!</h1>
<p>Aceasta este pagina servită manual din directorul <code>static/</code>.</p>
</body>
</html>
* Host localhost:8000 was resolved.
* IPv6: ::1
* IPv4: 127.0.0.1
% Total % Received % Xferd Average Speed Time Time Time Current
0 0 0 0 0 0 0 0 --:--:-- --:--:-- --:--:-- 0* Trying [::1]:80
00...
* connect to ::1 port 8000 from ::1 port 43520 failed: Connection refused
* Trying 127.0.0.1:8000...
* Connected to localhost (127.0.0.1) port 8000
> GET /doesnotexist HTTP/1.1
> Host: localhost:8000
> User-Agent: curl/8.5.0
> Accept: */*
>
< HTTP/1.1 404 Not Found
< Content-Type: text/html
< Content-Length: 100
< Connection: close
<
{ [100 bytes data]
100 100 100 100 0 0 43535 0 --:--:-- --:--:-- --:--:-- 50000
* Closing connection
<html><body><h1>404 Not Found</h1><p>Resursa solicitata nu exista pe acest server.</p></body></ht
ml>
--- PUNCTUL 5: EXPLICATIE TEHNICA ---
Serverul parseaza request-ul prin citirea datelor brute de pe socket si extragerea primei linii,
pe care o imparte in fragmente pentru a identifica metoda (GET) si calea (path). Decizia despre c
e fisier sa trimita se bazeaza pe concatenarea directorului 'static' cu calea extrasa din URL. An
tetul Content-Length este necesar pentru ca browserul sa stie exact cati bytes de date urmeaza sa
primeasca inainte de a inchide conexiunea. Cand un fisier nu exista pe disc, serverul prinde ero
area prin verificarea os.path.isfile si trimite inapoi un cod de stare 404 Not Found. Intreg proc
esul demonstreaza cum protocolul HTTP functioneaza ca un schimb de mesaje text structurate peste
TCP.
gr5_3@svs:~/s6S

```